

Optimal investment and restarting decisions during decommission for the Davy Oil Field

Oli O'Malley

September 2026

1 Abstract

This paper develops a continuous-time real options model to evaluate investment decisions and optimal restart triggers under uncertainty during asset decommissioning. The model utilizes a Markov-chain approximation finite differences scheme to find the boundary line which shows when a Decision Maker should repair and restart a project once decommissioning has begun. There is also then an optimal restart trigger calculated for the mothballing phase where a project is paused, but decommission is yet to begin. The Davy Oil Field is used and it is shown that it is almost never optimal to repair and restart once decommission has begun, which is reflected in the common real world decision to keep large oil rigs idle for as long as possible.

2 Introduction

This model is adapted from Thijssen (2022) where it is shown that large scale projects are almost always over budget and behind schedule. Whereas Thijssen (2022) evaluates the option to abandon a project during its construction phase, this paper logically inverts the framework to evaluate the optimal boundary for repairing and restarting an asset during its active decommissioning phase. This paper adapts his model to evaluate the specific empirical case of the Davy Oil Field, introducing a non-linear scaling repair cost function to simulate the option a DM has to repair and restart the Davy Oil Field once full scale dismantling has begun. This specific empirical case was selected because the field remained idle from 2021 to 2025 under a nominal mothballing fee, until commodity revenues rose sufficiently to justify reopening operations in 2026. This framework demonstrates this exact decision, the revenues required in order to consider restarting, and then the revenues required once you are forced to stop mothballing and begin dismantling.

3 The Set-up

The model uses stochastic processes for both decommissioning and revenues, with decommission costs being constant. The main parameters are as follows:

1. a (stochastic) decommission process $(X_t)_{t \geq 0}$, $X_0 = 0$;
2. a constant decommission cost $c > 0$;
3. a perpetual (stochastic) revenue flow $(Y_t)_{t \geq 0}$;
4. a perpetual constant operational cost flow $\eta > 0$;
5. a perpetual constant cost flow $\kappa \in [0, \eta)$, the mothball cost for when the project is idle but before decommission has begun;
6. a constant discount rate $r > 0$;
7. an exponential repair cost function $K_{repair}(x)$, that can be taken once decommission has begun ;

8. a sunk restart cost $K_{restart}$, that is taken to restart during mothball phase, and also after during decommission phase alongside the repair function, if revenues are high enough;
9. a net convenience yield $\delta = r - \mu_2$ which is positive at the current time.

As is standard in the real options literature, the stochastic processes governing this model are Markovian. We define uncertainty on a state space $E = [0, 100] \times (0, \infty)$, where x represents the percentage of dismantlement progress and y represents the annual revenues from the operational Davy gas field.

On a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with a natural filtration $\mathcal{F} = (\mathcal{F}_t)_{t \geq 0}$, we define two correlated standard Brownian motions B^X and B^Y , such that $\mathbb{E}[dB^X dB^Y] = \rho dt$ for some correlation coefficient $\rho \in (-1, 1)$. The dismantlement progress $X = (X_t)_{t \geq 0}$ follows an arithmetic Brownian motion with drift, reflecting the forward trajectory of decommissioning alongside operational uncertainties (e.g., weather delays):

$$dX_t = \mu_1 dt + \sigma_1 dB_t^X$$

The revenue process $Y = (Y_t)_{t \geq 0}$ follows a standard Geometric Brownian Motion:

$$dY_t = \mu_2 Y_t dt + \sigma_2 Y_t dB_t^Y$$

where $\mu_1, \mu_2, \sigma_1, \sigma_2 > 0$.

We assume that the stochastic processes governing decommissioning progress and asset revenue are independent ($\rho = 0$). This reflects the empirical reality that the localized operational and weather risks that affect the dismantling of the rig (X) are fundamentally uncorrelated with the global macroeconomic factors driving natural gas prices (Y).

4 The Model

We begin with the value of the project during the decommissioning phase:

$$F(x, y) := \sup_{\tau \leq \tau_*} \mathbb{E} \left[-c \int_0^\tau e^{-rt} dt + e^{-r\tau} (\mathbf{1}_{\tau \leq \tau_*} (V_{operate}(y) - K_{repair}(x) - K_{restart}) + \mathbf{1}_{\tau = \tau_*} (0)) \right] \quad (1)$$

Where repair is modeled as a scaled exponential function, to simulate how there would be more to repair the further the dismantlement progresses.

$$V_{operate}(y) = \frac{Y_\tau}{\delta} - \frac{\eta}{r} \quad (2)$$

$$K_{repair}(x) = C(e^{mx} - 1) \quad (3)$$

There is no known analytical solution to (1), the next section solves this numerically with a markov chain approximation. Because the capital outlay for dismantling strictly dominates the operational cash flows, a voluntary trigger to initiate decommissioning does not exist. Therefore, the primary focus of this model is identifying the optimal boundary at which operating revenues justify reversing a forced decommissioning decision.

Evaluating the preceding mothball phase, the decision to restart production constitutes a standard optimal stopping problem. Because the trigger to begin decommissioning is regulatory rather than voluntary, the system simplifies to standard value-matching and smooth-pasting conditions: So the value of waiting must equal the value of restarting, and the rate of change of waiting must equal that of restarting:

$$Ay^{\beta_1} - \frac{\kappa}{r} = \frac{y}{r - \mu_2} - \frac{\eta}{r} - K_{restart} \quad (4)$$

$$A\beta_1 y^{\beta_1 - 1} = \frac{1}{r - \mu_2} \quad (5)$$

Which can be solved analytically and yields the trigger:

$$y^* = \frac{\beta_1}{\beta_1 - 1} (r - \mu_2) \left(\frac{\eta - \kappa}{r} + K_{restart} \right) \quad (6)$$

5 The Markov Chain Approximation

Since there is no analytical solution to (1), a Markov Chain approximation scheme is used, a numerical method as seen in Kushner and Dupuis (2001). This involves discretizing the space and solving a HJB equation at each node in order to approximate the value function across the entire space. First, the value function requires a log transform $z := \log(y)$, creating a grid $\mathcal{G} \triangleq [\underline{x}, \bar{x}] \times [\underline{z}, \bar{z}]$, with $h = 0.1$ as the step size of the grid. In particular, $z_{min} = -10.0$ up to $z_{max} = 7.70$ and $x_{min} = -100$ to $x_{max} = 100$ is sufficient and creates a 178x2001 grid. The log transform also ensures the transition probabilities to be constant across the grid.

$$\hat{F}(x, z) := \sup_{\tau \leq \tau_*} \mathbb{E} \left[-c \int_0^\tau e^{-rt} dt + e^{-r\tau} (\mathbf{1}_{\tau \leq \tau_*} (V_{operate}(e^z) - K_{repair}(x) - K_{restart}) + \mathbf{1}_{\tau = \tau_*}(0)) \right] \quad (7)$$

The repair function boundary condition is set to $K_{repair}(0) = 0$, reflecting that an intact asset requires no repair. The scaling constants are calibrated to empirical estimates, assuming a baseline £120 million total replacement cost and an initial £2 million repair floor. This yields the final repair function:

$$K_{repair}(x) = 2(e^{4.11x} - 1) \quad (8)$$

We then set up the HJB equation :

$$\begin{aligned} \hat{F}(x, z) = \max \{ & \mathbf{1}_{\{x < x^*\}} (V_{operate}(e^z) - K_{repair}(x) - K_{restart}) + \mathbf{1}_{\{x = x^*\}}(0), \\ & \hat{\mathcal{L}}\hat{F}(x, z) + (1 - r)\hat{F}(x, z) - c \} \end{aligned} \quad (9)$$

Where $\hat{\mathcal{L}}$ is the characteristic operator of (X,Z). We then, as shown in Thijssen(2022), sub in the finite difference approximations and rearrange to get the transition probabilities. These importantly sum to 1 in order for stability in the grid to ensure it converges to a true stable answer. The HJB equation (8) is then replaced by the discrete-time approximation

$$\hat{F}^h(x, z) = (T^h \hat{F}^h)(x, z) \quad (10)$$

Where the operator T is given by

$$\begin{aligned} (T^h \hat{F}^h)(x, z) := \max \{ & \mathbf{1}_{\{x < x^*\}} (V_{operate}(e^z) - K_{repair}(x) - K_{restart}) + \mathbf{1}_{\{x = x^*\}}(0), \\ & (1 - r\Delta t^h(x, z))\hat{\mathcal{L}}^h \hat{F}^h(x, z) - c\Delta t^h(x, z) \} \end{aligned} \quad (11)$$

In order to run the scheme, the following boundary conditions are set up.

1. During decommission, if revenues are excessively high, there is almost certainty that the project would be restarted. That is $\hat{F}(x, \bar{z}) = V_{operate}(e^{\bar{z}}) - K_{repair}(x) - K_{restart}$
2. Once decommission has finished, the value of the project is worth nothing as it is completely dismantled. That is $\hat{F}(\bar{x}, z) = 0$
3. When revenues are very low, there is almost no chance of restarting, so its known for certain that the project will reach full dismantlement and give 0 reward plus the present value of the cost stream of decommission: $\hat{F}(x, \underline{z}) = 0 - \frac{c}{r} (1 - e^{\alpha_1(x - x_{max})})$

The numerical scheme is solved iteratively by applying the operator T , continuing until the uniform convergence between F_n^h and $\hat{F}_{n-1}^h$ satisfies the predefined tolerance threshold.

6 Parameters

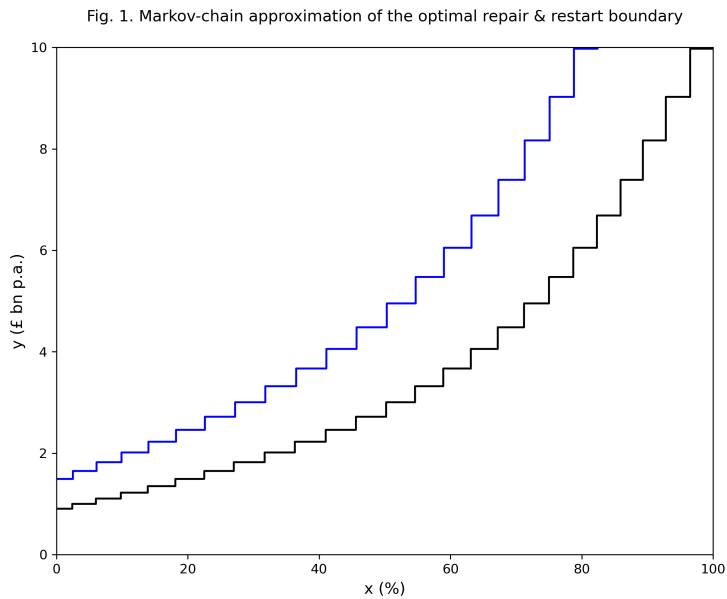
Here are the parameters I've found/ made assumptions for the Davy Oil Field

Symbol	Description	Value	Source
r	Discount rate (continuous)	0.077	assumed
μ_1	Dismantling progress drift (% p.a.)	150	assumed
σ_1	Dismantling progress volatility (%)	25	assumed
μ_2	Expected revenue growth rate	0.015	assumed
σ_2	Revenue volatility	0.45	inferred
c	Dismantling costs (£m p.a.)	54.75	inferred
η	Operating costs (£m p.a.)	3.65	inferred
k	Mothball holding costs (£m p.a.)	0.73	inferred
K_{restart}	Rig restart cost (£m)	2.00	inferred
x^*	Project completion target (%)	100	assumed
δ	Net Convenience Yield (%)	10.2	assumed

Table 1: Base case parameters for the Davy field valuation and numerical grid.

7 Findings

Model simulations indicate that to completely cancel the dismantling phase at $x = 0\%$, baseline revenues must reach an prohibitive threshold of £904.84 million p.a. as shown by the black line in Fig. 1. Given that initial operating revenues stand at approximately £8.5 million p.a., this would suggest very little value in considering restarting. This is driven primarily by the severe cost of the dismantling, approximating £150,000 per day. The trigger for active dismantling during the mothball phase does not exist. Because incurring the £54.75 million p.a. dismantling cost is strictly dominated by paying the nominal £0.73 million p.a. mothball holding fee, voluntary decommissioning is strictly sub optimal. In practice, operators only initiate physical decommissioning under strict regulatory compulsion. Conversely, applying standard value-matching and smooth-pasting conditions yields a viable restart threshold during the mothball phase, placing the optimal revenue trigger at £8.72 million p.a. This output aligns closely with historical precedent, as elevated commodity prices justified the operational restart of the asset. Incorporating a 10.2% net convenience yield to capture prevailing market backwardation shifts the value boundaries upward. As illustrated by the blue line in Fig. 1, the threshold to cancel demolition at $x = 0\%$ rises to £1,491.82 million, while the mothball-to-operation restart trigger increases to £9.93 million.



8 References

- Thijssen, J. J. (2022). Optimal investment and abandonment decisions for projects with construction uncertainty. *European Journal of Operational Research*, 298(1), 368-379.
- Kushner, H. J., & Dupuis, P. G. (2001). *Numerical methods for stochastic control problems in continuous time*. Springer Science & Business Media.